



EVN NPT

The National Power Transmission Corporation

APPLICATION OF ARTIFICIAL INTELLIGENCE IN IMAGE PROCESSING FOR POWER TRANSMISSION LINE INSPECTION

February, 2025



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October 2024



EVNNPT

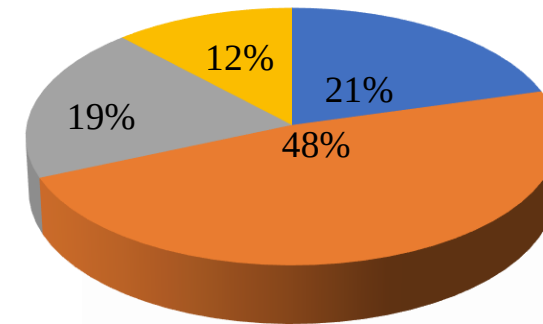
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1. Transmission lines scope

✓ EVNNPT manages and operates (As of 30/09/24) more than 31,500 km (circuits length) of 500kV, 220kV transmission lines across complex terrains, ranging from high mountainous areas to low plain field.

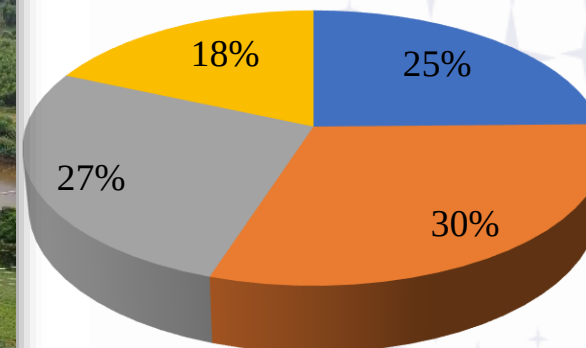


500kV lines



- Area especially difficult, harsh climate, lots of thunderstorms
- High mountains, dense forests, swamps

220kV lines



- Midlands, sparse forests, inner cities, coastal areas
- Plain



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2. Methods of transmission lines management

- Before 2019: workers patrol along transmission line for visual inspection and taking photos of any abnormalities.
- From 2019 to 2022: unmanned aerial vehicles (drones) and cameras installed on electric poles had been using in operational management.
- From 2022: artificial intelligence has been applied to support image analysis and equipment condition assessment.



2. Methods of transmission lines management



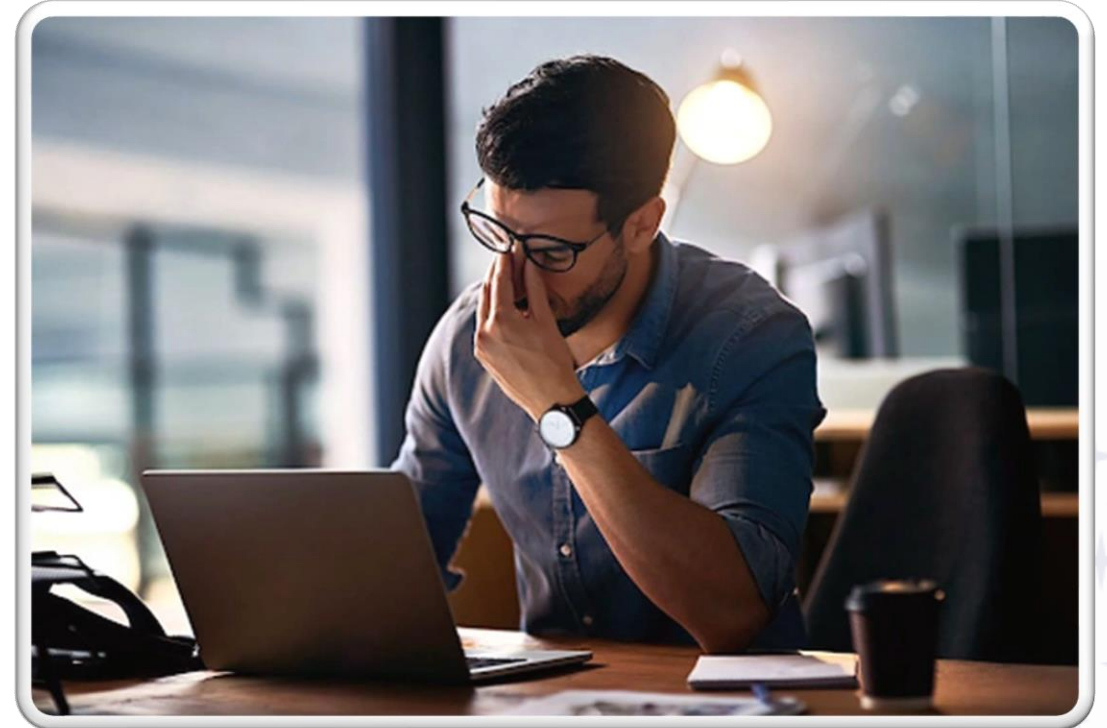
Before:
Inspection and Assessment of transmission lines – Paper based

Present:
Inspection of transmission lines by UAV to capture image
Assessment of transmission lines supported by AI for image processing

2. Methods of transmission lines management

Why we choose AI:

- Images captured from drones taken time to edit, arrange and store manually on storage devices.
- Images are checked manually by our employee and it could take long time for finish, hugely depending on capability of our inspector



2. Methods of transmission lines management

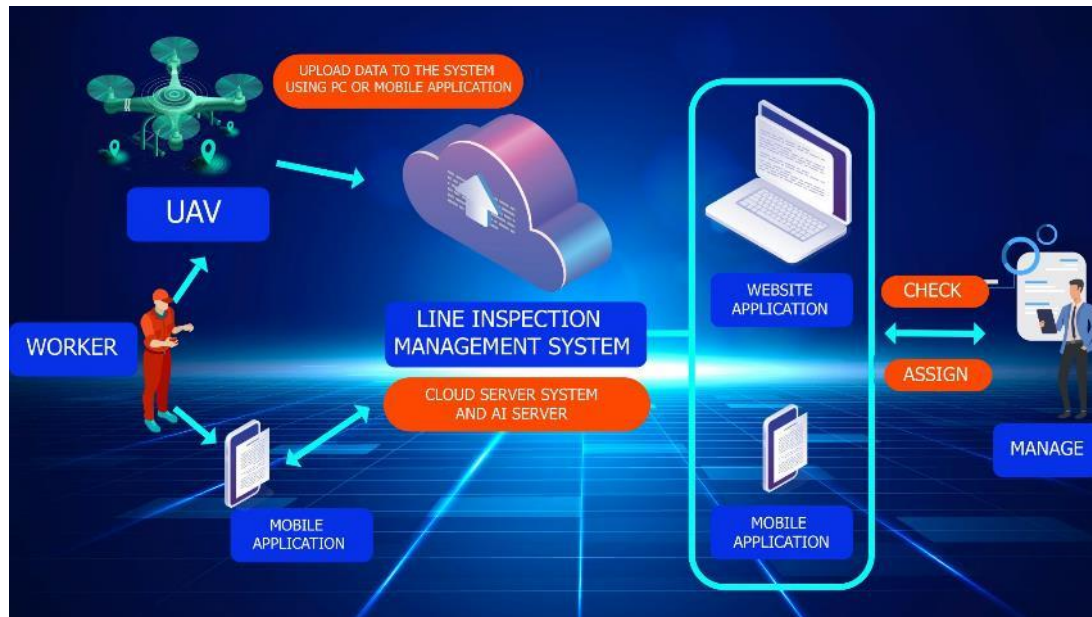
Aims of our solution

- Automatically organize, classify, store data (images), and manage information after each inspection.
- Automatically check and detect defects and damage through data (images) collected from recording devices. Provide preliminary assessments so that workers can recognize quickly and focus on checking, confirming, and handling arising problems. Gradually replace operators in image inspection work.

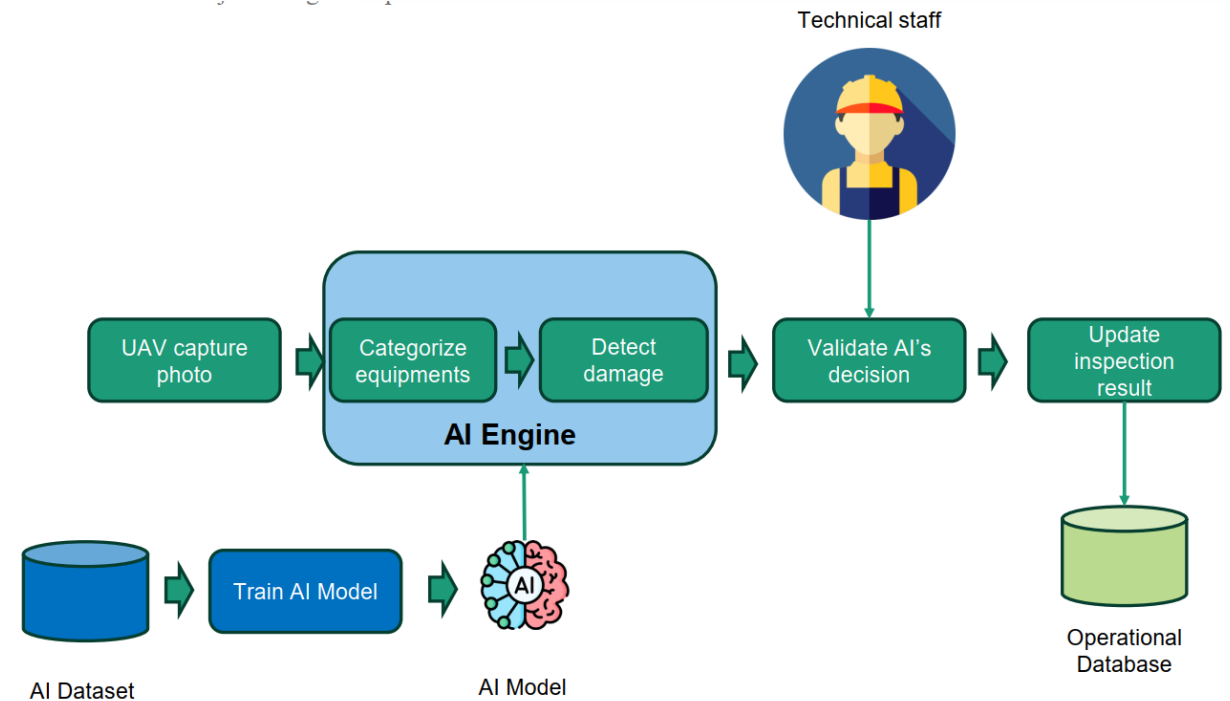


2. Methods of transmission lines management

Our solution



Our partner: 

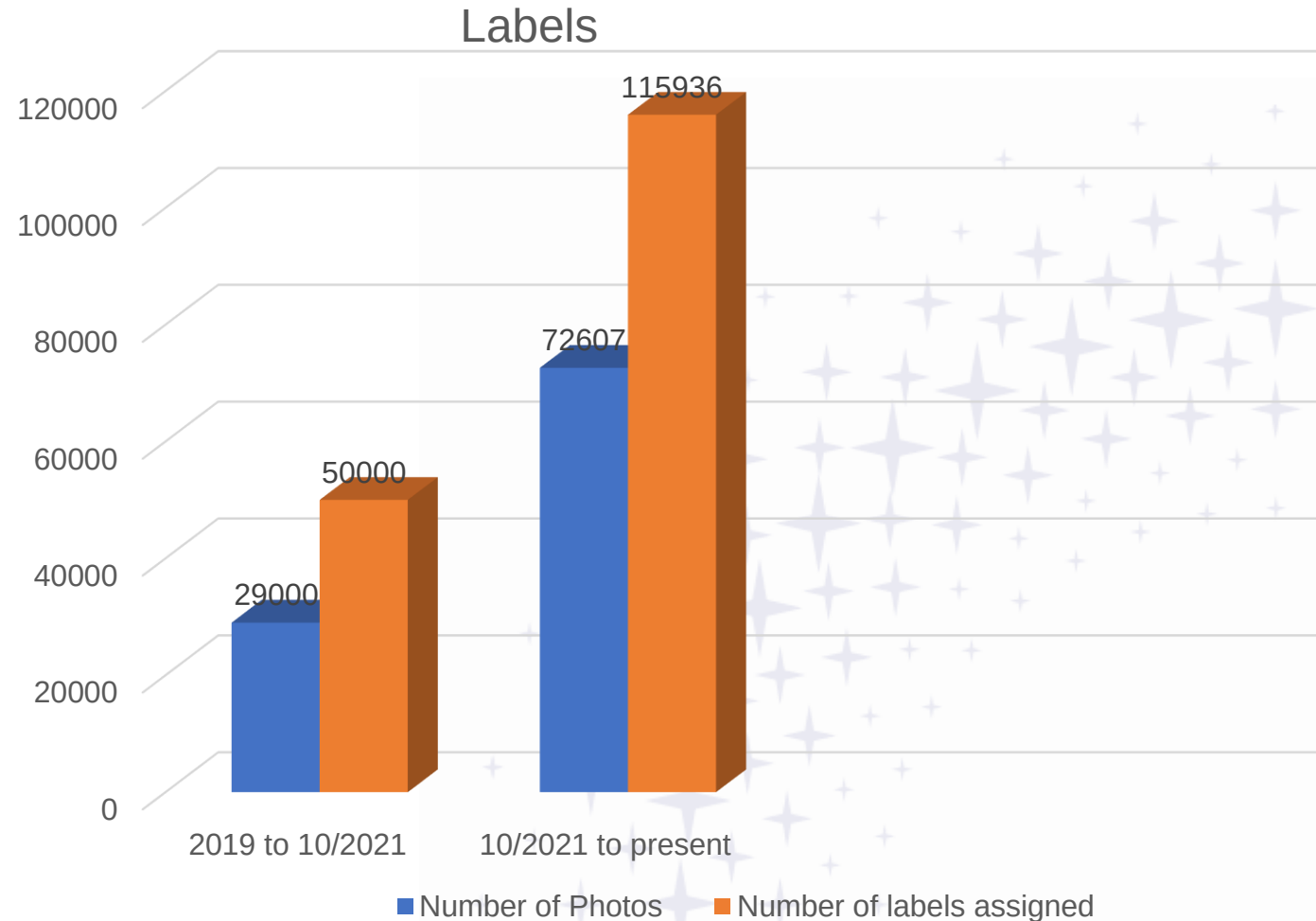


3. Research and deployment of AI models:

3.1. Image Analysis and Checking by Artificial intelligence (AI) System

- EVNNPT/PTC2 has built a platform for labeling, training AI models, and creating APIs for other management software.
- Performing data collection and labeling for 72,607 photos with 115,936 labels of equipment and accessories on the line, including:
 - + 19 normal labels
 - + 07 abnormal labels

However, there are some labels that are difficult to recognize in regular drone images.





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3. Research and deployment of AI models:

3.1. *Image Analysis and Checking by Artificial intelligence (AI) System*

Difficulties in Performing Image Collection and Labeling:

- A total of 110 power transmission units must be mobilized for data collection and labeling.
- The process requires significant efforts and time.
- Abnormal labels are often scarce, making collection very challenging.





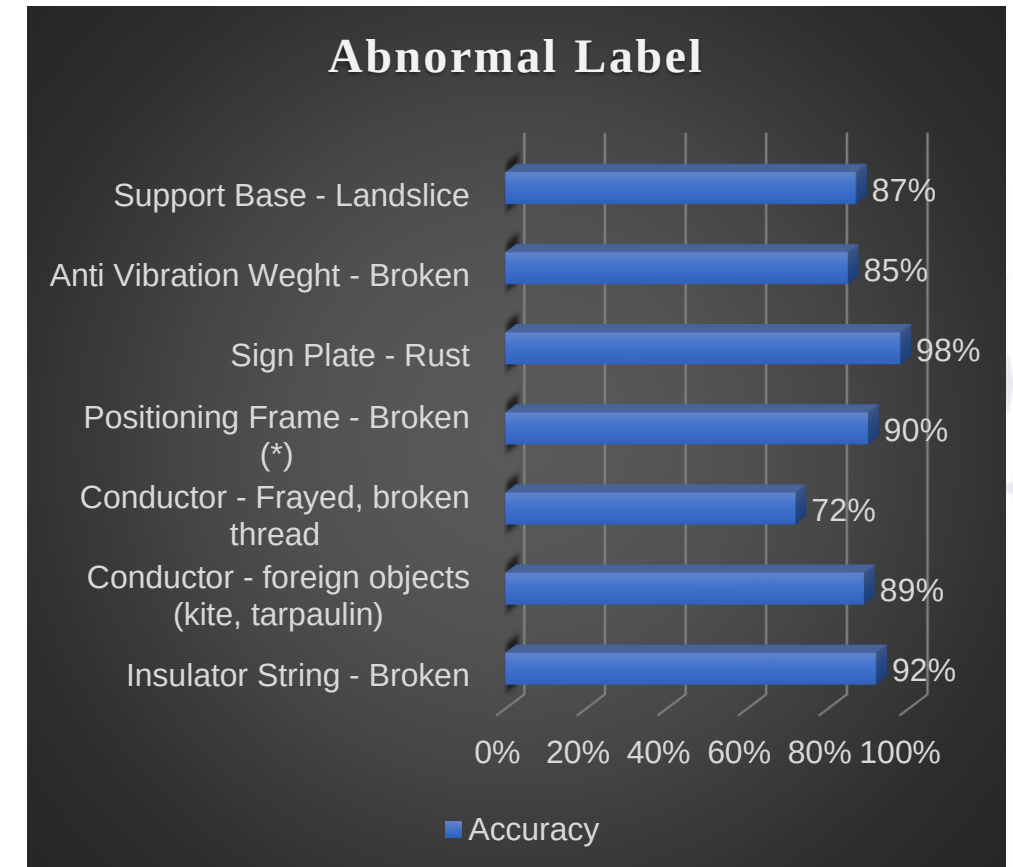
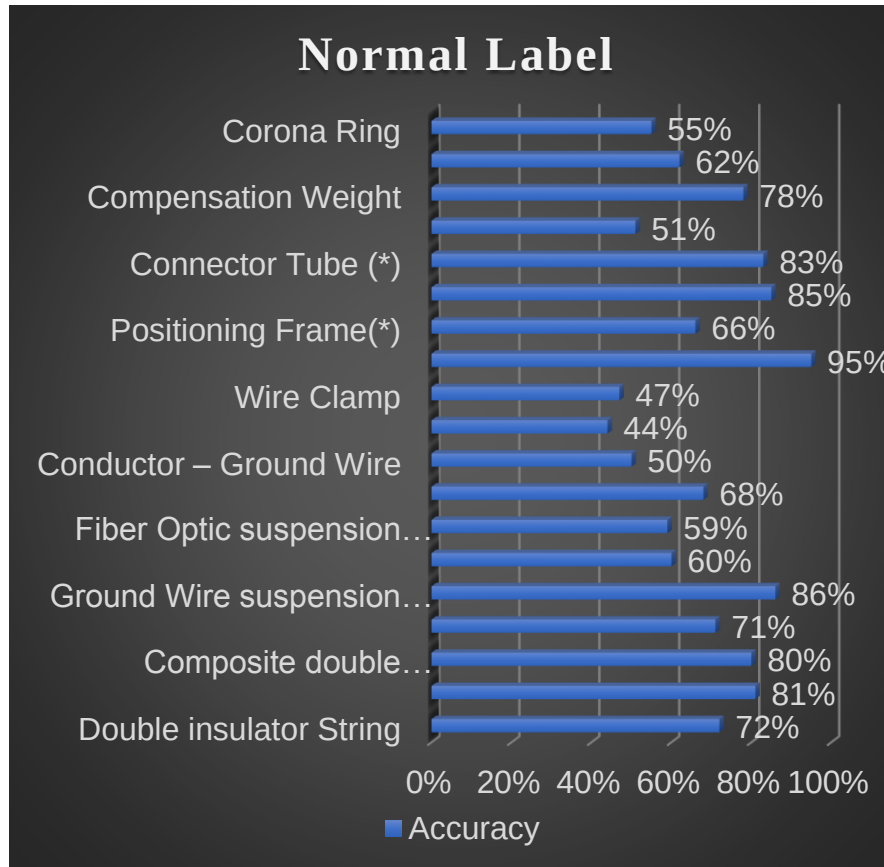
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3. Research and deployment of AI models:

3.1. Image Analysis and Checking by Artificial intelligence (AI) System

*AI model
accuracy*



(*) These labels are difficult to identify with regular, remote drone images.

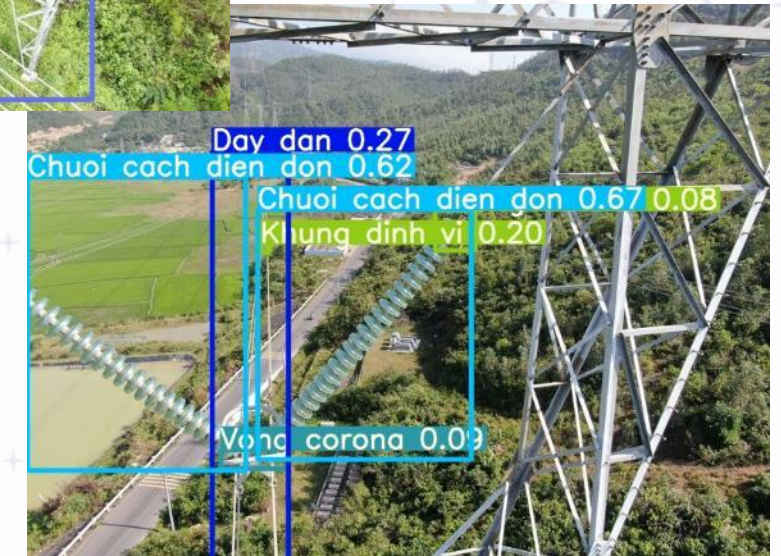


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3. Research and deployment of AI models

3.1. Image Analysis and Checking by Artificial intelligence (AI) System



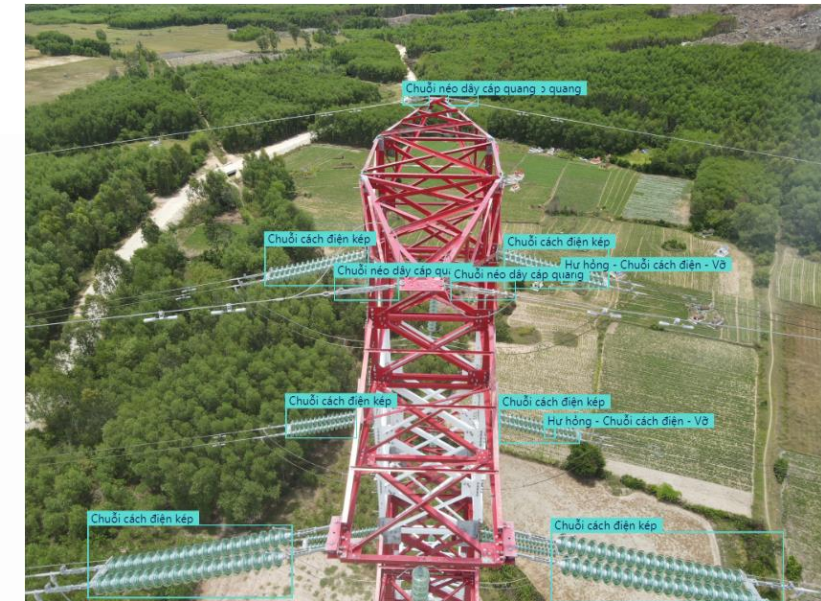
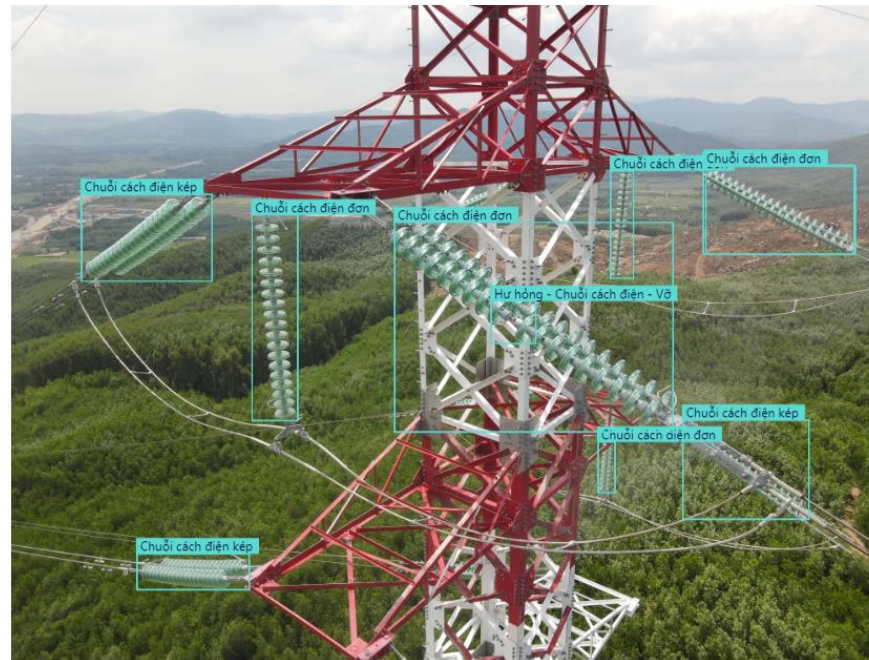
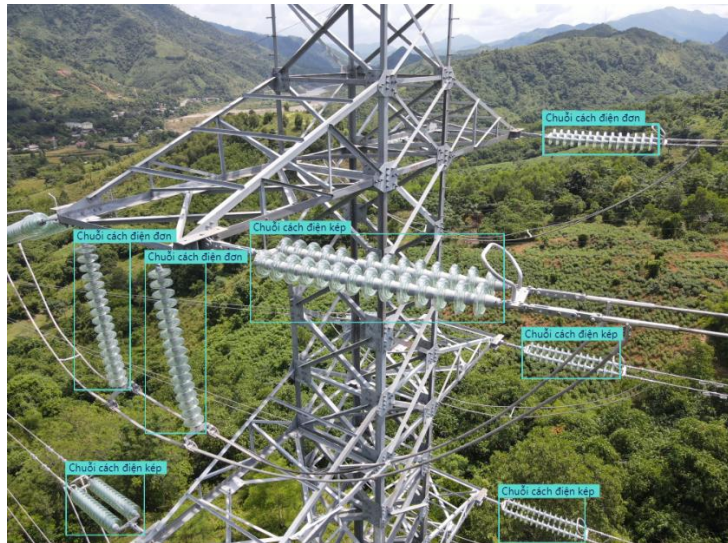


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3. Research and deployment of AI models

3.1. Image Analysis and Checking by Artificial intelligence (AI) System



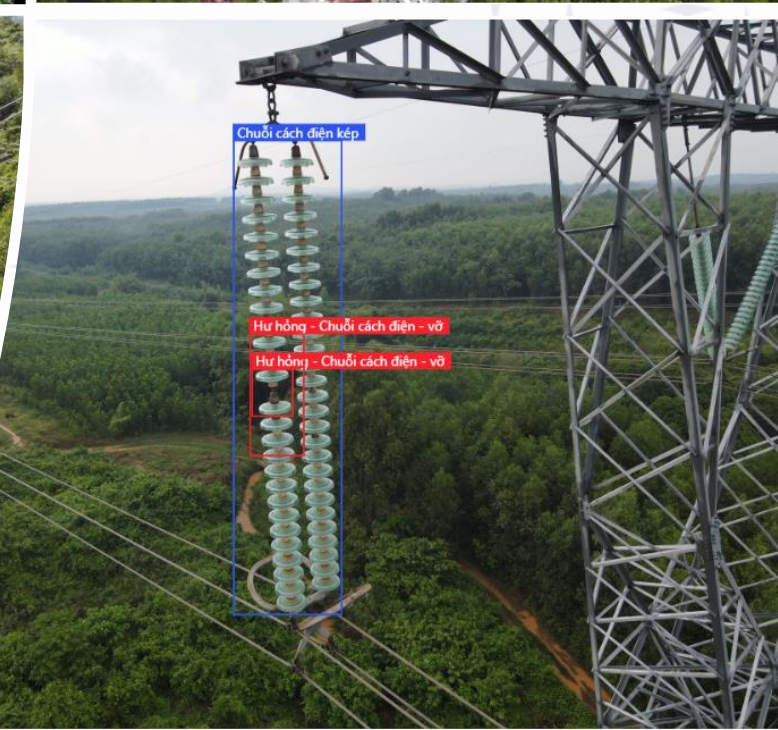
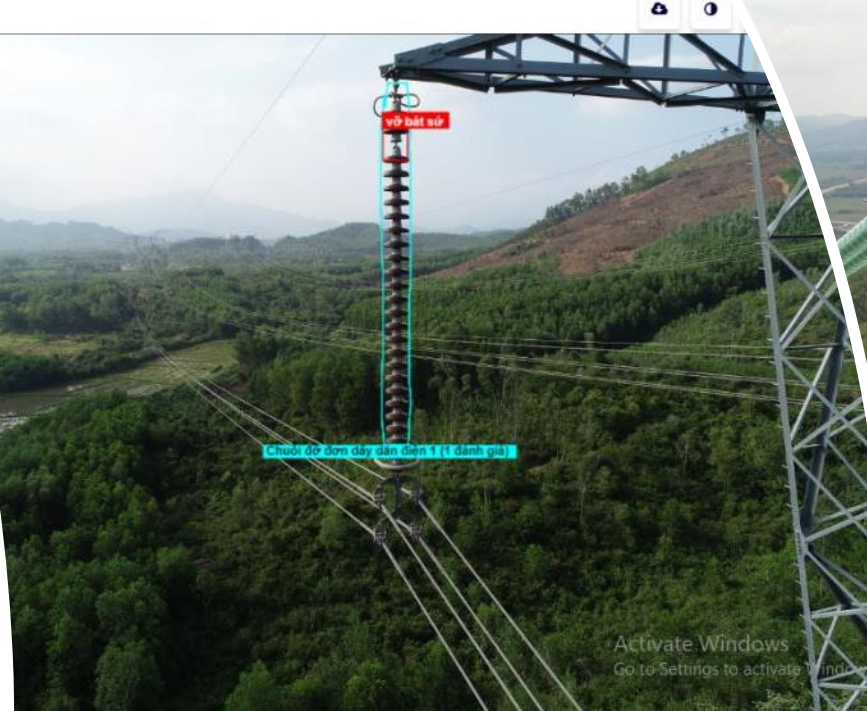


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3. Research and deployment of AI models

3.1. Image Analysis and Checking by Artificial intelligence (AI) System



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3. Research and deployment of AI models

3.1. Image Analysis and Checking by Artificial intelligence (AI) System

AI is the core solution to improve the quality and efficiency of line management and increase labor productivity, and is the driving force to shift line management and operation labor from traditional to modernized labor

Efficiency:

- *Automatically, quickly and effectively detect defects/damages using bulk collected data.*
- *Reduce time and effort when checking large amounts of data on the computer.*
- *Minimize the feeling of boredom and fatigue for the inspector when checking data.*

Limitations:

- *In order to ensure the required accuracy for the recognition system, it is necessary to build a very large database through data collection (images) and label the images. Therefore, it will take a lot of time and labors.*
- *The ability of identifying accurately is high but the damage and abnormalities are relatively large and have to be visible for visual inspection.*
- *The model will be noisy when being trained if the number of labels is not similar (the difference is more than 500 labels)*

3. Research and deployment of AI models

3.2. *Line surveillance cameras based Artificial intelligence (AI) system*

We have deployed 352 camera surveillance systems on the lines, most of which have been integrated with the AI system. This AI model will:

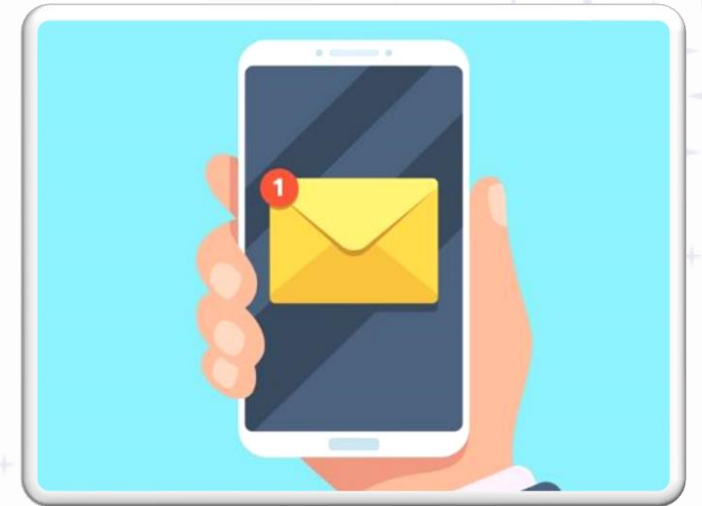
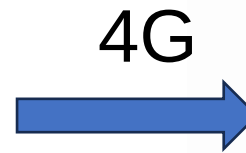
- Detect abnormal detections on the line including: fire, smoke and motor vehicles near the line corridor.
- Labels and trains with over 8,000 labels for the above abnormalities.
- Be deployed and installed on most of our 500kV and 200kV lines.



+



Computer with AI



3. Research and deployment of AI models

3.2. Line surveillance cameras based Artificial intelligence (AI) system



Có 1 phương tiện/đám cháy được phát hiện tại camera VT99 ĐZ500kV Dốc Sỏi – Pleiku 2 vào lúc 10:40:51 ngày 15/03/2022. Đề nghị đơn vị

There was a fire detected at camera VT99 ĐZ500kV Doc Soi - Pleiku 2 at 10:40:51 on March 15, 2022. We recommended that the operating management unit checked the camera and handled it.



Có 1 phương tiện/đám cháy được phát hiện tại camera VT 13 ĐZ500kV Đông Hà – Huế vào lúc 08:46:06 ngày 12/10/2024. Đề nghị đơn vị

There was a vehicle detected at camera VT13 ĐZ500kV Dong Ha - Hue at 08:46:06 on October 12, 2024. We recommended that the operating management unit checked the camera and handled it.



Có 1 phương tiện/đám cháy được phát hiện tại camera VT 102 ĐZ500kV XKM 3 – Thanh Mỹ 500 vào lúc 13:33:20 ngày 07/06/2022.

There was a fire detected at camera VT102 ĐZ500kV XKM3 – Thanh My 500 at 13:33:20 on June 7, 2022. We recommended that the operating management unit checked the camera and handled it.



Có 1 phương tiện/đám cháy được phát hiện tại camera VT 102 ĐZ500kV XKM 3 – Thanh Mỹ 500 vào lúc 15:13:47 ngày 11/06/2022.

There was a fire detected at camera VT102 ĐZ500kV XKM3 – Thanh My 500 at 15:13:47 on June 11, 2022. We recommended that the operating management unit checked the camera and handled it.



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3. Research and deployment of AI models

3.2. *Line surveillance cameras based Artificial intelligence system*

***Effectiveness:**

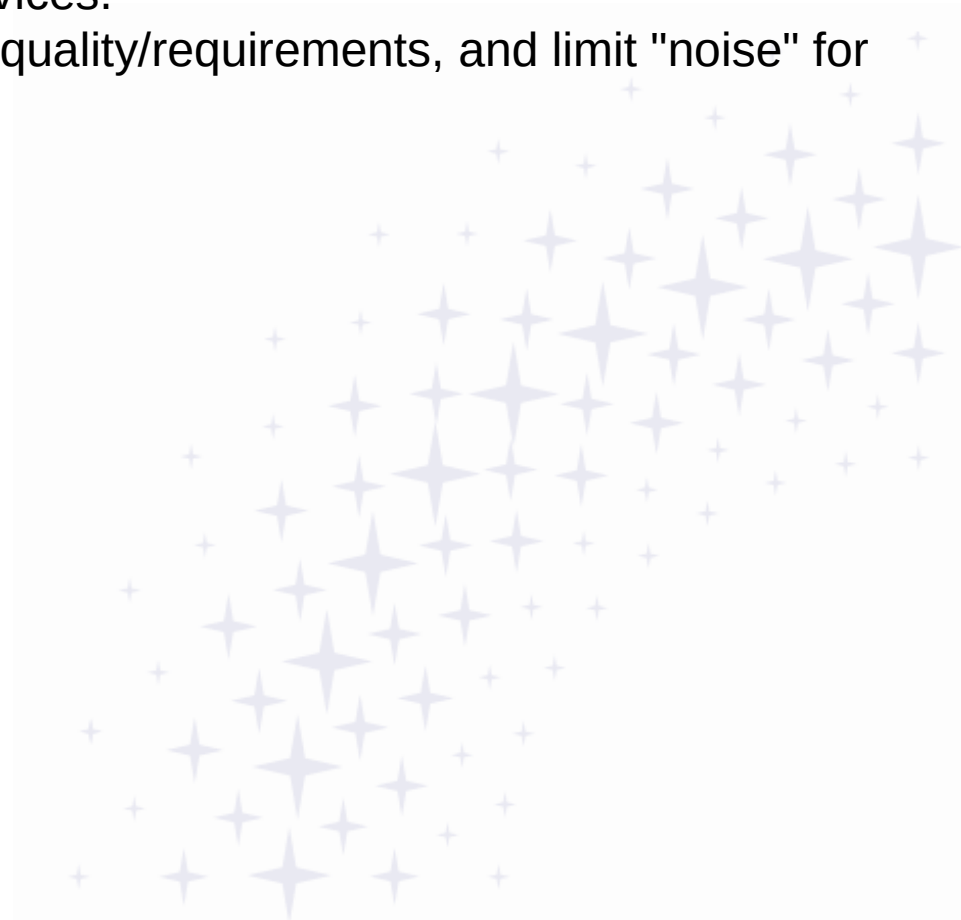
- Helps automatically warn of some risks that cause unsafety during power grid operation.
- Supports our employees in monitoring the grid instead of sending people for site inspection or continuously monitor on the camera system.
- Helps optimize human resources, means, and improves production efficiency.

*** Limitations:**

- The AI model does not really work accurately with similar objects (clouds) and recognize activities at night.
- The AI system on this camera is researched and developed by EVNNPT/PTC2, so there is no centralized management system for warning data at the present
- Problems related to the power supply of the camera system and AI happening, including the errors/damages of the solar panel controller, trouble with the 4G router.

4. Future work

- Deploy the AI model for all transmission line unit under the National Power Transmission Corporation.
- Strengthen the work of labeling and training AI models to improve the accuracy of the AI system in identifying devices as well as abnormal phenomena and incidents related to devices.
- Review and standardize labeled data to limit labels that do not meet quality/requirements, and limit "noise" for the AI system.
- Apply AI model for substation management.



Thank You!

